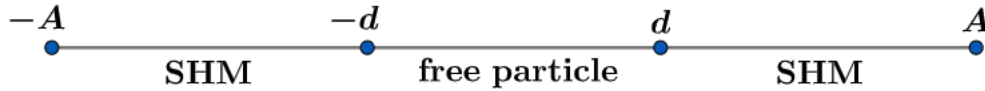


2021 F=ma Exam: Problem 23

Kevin S. Huang



Recall that the period for a mass-spring system is given by

$$T_0 = 2\pi\sqrt{\frac{m}{k}}$$

We need to add to this the time the particle moves between d and $-d$ during each period. The amplitude of the simple harmonic motion is $A - d$ so by conservation of energy,

$$\begin{aligned}\frac{1}{2}mv^2 &= \frac{1}{2}k(A - d)^2 \\ v &= (A - d)\sqrt{\frac{k}{m}}\end{aligned}$$

The time the particle takes to go across the region with no forces is

$$t = \frac{2d}{v} = \frac{2d}{A - d}\sqrt{\frac{m}{k}}$$

This happens twice every period of oscillation so

$$T = T_0 + 2t = \sqrt{\frac{m}{k}}\left(2\pi + \frac{4d}{A - d}\right)$$

Thus, the answer is $\boxed{\text{C}}$.